

Finding a Shorter Side

Pythagoras' Theorem. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The learner finds an unknown *shorter* side of a right-angled triangle, given the hypotenuse and the other shorter side. The key shift from the previous lesson is direction: because the hypotenuse's square is the largest, the method subtracts — $a^2 = c^2 - b^2$ — before the final square root.

Driving Question: *How can I use the hypotenuse and one shorter side to find the missing shorter side?*

The mathematics, in plain terms

Pythagoras' Theorem, $a^2 + b^2 = c^2$, can be rearranged: a shorter side's square equals the hypotenuse's square minus the other shorter side's square ($a^2 = c^2 - b^2$). The steps: identify the hypotenuse (largest square); square the hypotenuse and the known shorter side; subtract the smaller from the larger; take the square root. Because we work back from the largest square, we take one away — the opposite direction to finding the hypotenuse.

Why we subtract here, and in which direction

The single most important idea is that the hypotenuse's square is the biggest of the three. So when a shorter side is missing, its square is what is *left* after taking the known shorter square away from the hypotenuse square. Subtracting the wrong way round (known minus hypotenuse) gives a negative area — a useful signal that the hypotenuse has been mis-identified. Encourage the learner to name the hypotenuse first, every time.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The missing side connects to the hypotenuse and the known side through the same theorem, now rearranged.	~8 min
Movement	Working the method backwards — from the largest square, taking one away.	~22 min
Reflection	Checking direction: am I subtracting from the largest square? Is my answer shorter than the hypotenuse?	~4 min
Creativity	Making a worked example that shows the subtraction clearly.	~3 min
Rest	The pause to decide add or subtract before diving in.	~3 min

Nutrition	Intellectual (Mode B): feeds careful decision-making and directional reasoning.	~5 min
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Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Adding instead of subtracting

The learner reaches for $a^2 + b^2$. Repair: “Is the side you want the longest one, or a shorter one? If it is shorter, are you building up to the biggest square, or taking away from it?”

Subtracting the wrong way round

The learner computes known square minus hypotenuse square. Repair: “Which square is the largest? We take the smaller away from the larger — which is which here?”

Forgetting the square root

Repair: “You have found the area of the square on the missing side. What length gives a square of that area?”

Mis-identifying the hypotenuse, so subtracting from the wrong square

Repair: “Point to the right angle. The hypotenuse is opposite it and carries the largest square. Are you subtracting from that one?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: square numbers, square roots, and recognising that the hypotenuse carries the largest square. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Keep the decision — add or subtract — with the learner:

- “Which side are we looking for — the longest, or a shorter one?”

- “Which square is the biggest? Are we adding to it, or taking away from it?”
- “Square the hypotenuse. Square the side you know. Now which do you take from which?”
- “Is your answer shorter than the hypotenuse? Should it be?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“Why is this different from the last lesson?”

Last time the missing side was the hypotenuse, so we added two squares. Here the missing side is a shorter one, so we take a square away from the hypotenuse’s square. Same theorem, opposite direction.

“They got a negative number — what happened?”

Almost always it means the hypotenuse was mis-identified and the subtraction went the wrong way. Ask which side is opposite the right angle.

“Does the answer have to be a whole number?”

No — a sensible rounded decimal, with the working shown, is a strong answer.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Emerging → Developing → Secure → Mastering across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Emerging	Beginning to engage; names some features and makes a start with support.
Developing	Carries out the method with prompts; can explain part of the reasoning.
Secure	Works through independently and explains why each step is true.
Mastering	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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